

Capstone Proposal

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News Summariser

Overview :

The News Analyzer is a fully serverless platform. A user hits a public HTTPS endpoint on API Gateway, which triggers a Lambda function running under LabRole. The Lambda reads the NewsAPI key from Secrets Manager at runtime, fetches top headlines, stores results in DynamoDB, and returns the articles to the caller. CloudWatch captures logs and metrics.

Services used: API Gateway · Lambda · Secrets Manager · DynamoDB · CloudWatch · NewsAPI

Decision 1 — Compute

Chose Lambda over ECS Fargate.

Fargate offers more control and longer runtimes but charges per vCPU/memory hour even when idle. Lambda costs nothing between requests and fits a bursty, on-demand news fetch perfectly.

Pillar: Cost

Decision 2 — Database

Chose DynamoDB (PAY_PER_REQUEST) over RDS.

RDS wins on complex ad-hoc queries but charges per hour regardless of usage. DynamoDB on-demand has zero idle cost, which matches the low-traffic nature of this project.

Pillar: Cost

Decision 3 — Secret Management

Chose Secrets Manager over Lambda environment variables.

Environment variables are simpler but the API key is visible in plain text in the Lambda console and can be leaked via logs. Secrets Manager encrypts the value at rest and requires an explicit SDK call to retrieve it, keeping the key out of code and config entirely.

Pillar: Security

Review Findings & Responses

Finding	Response
No authentication on API Gateway endpoint	Accepted — will add an API key or IAM auth in a later session
Articles not persisted to DynamoDB yet	Accepted — fixing this now (see commit)
No error handling if NewsAPI is down	Accepted — will wrap in try/except with fallback response

Architecture Diagram :

